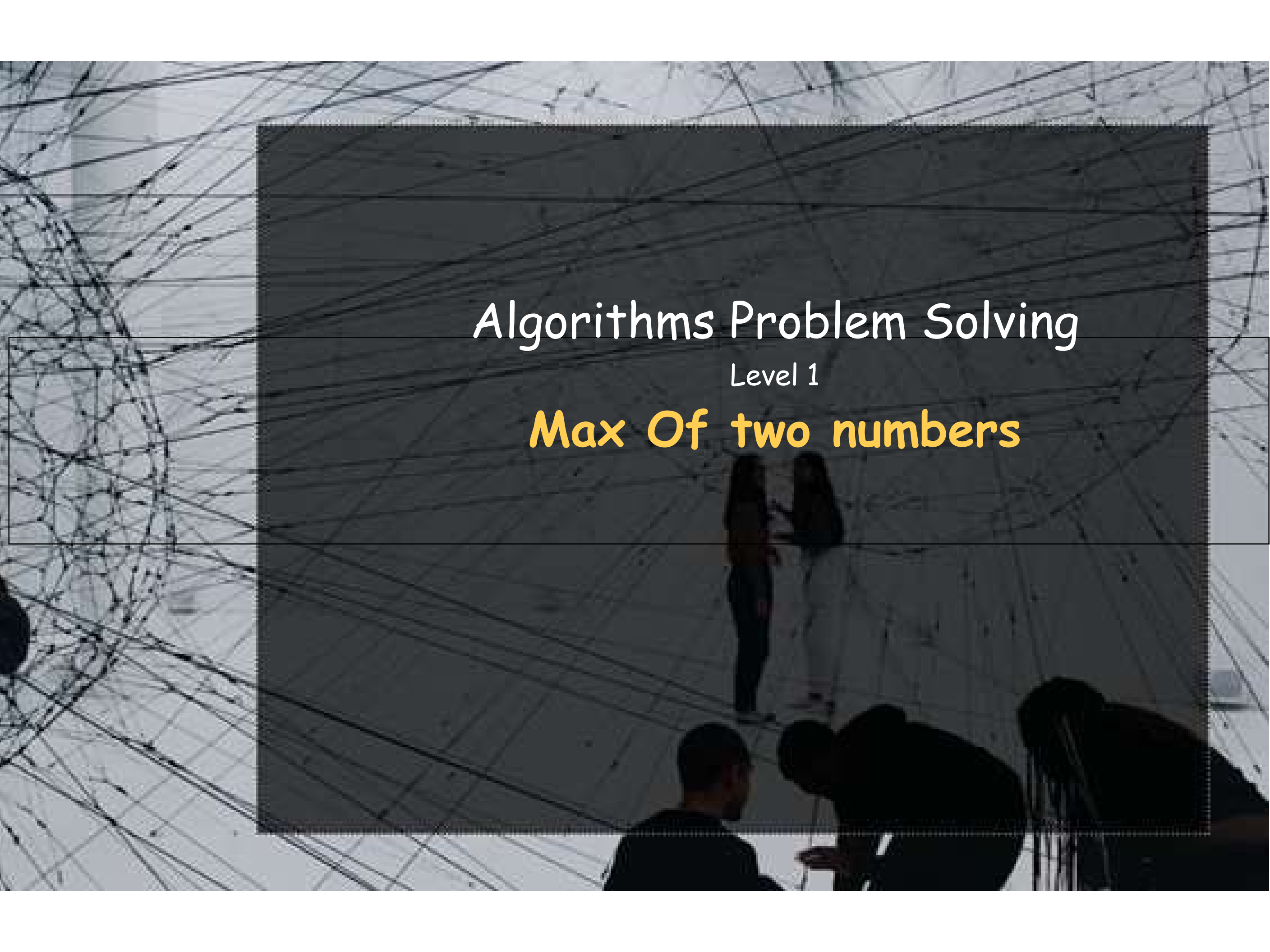


Algorithms Problem Solving

Level 1

Max Of two numbers



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Definition

- ◆ This program reads two numbers from the user and finds the maximum value between them.
- ◆ The program uses functions, reference parameters, an if-else condition, and a return value.

Explanation

- ◆ The `readNumbers()` function reads two numbers and stores them in `num1` and `num2`.
- ◆ The parameters `int &num1` and `int &num2` are reference parameters, so the function can directly modify the variables from `main()`.
- ◆ The `maxOf2Numbers()` function compares the two numbers:
- ◆ If `num1 > num2`, it returns `num1`.
- ◆ Otherwise, it returns `num2`.
- ◆ The `printResult()` function receives the maximum number and prints it.
- ◆ In `main()`, the functions are combined:
- ◆ `readNumbers(num1, num2);`
- ◆ `printResult(maxOf2Numbers(num1, num2));`

Steps in main()

1. Create two integer variables.
2. Read the first number using readNumbers().
3. Read the second number using readNumbers().
4. Compare the two numbers.
5. Return the greater number.
6. Pass the returned value to printResult().
7. Print the maximum number.

💡 Example

◆ If the user enters:

Number 1: 25

Number 2: 40

◆ The program compares 25 and 40.

◆ Since 40 is greater, `maxOf2Numbers()` returns 40.

➡ Result

The maximum number is 40

Note

- ◆ The & symbol in the parameters makes num1 and num2 reference parameters.
- ◆ This allows readNumbers() to change the original variables declared in main().

⊘ Warning

◆ If the two numbers are equal, the else statement returns either number because both have the same value.



Comparison

▨ num1 > num2

- ✓ num1 is returned.
- ✓ num1 is the maximum number.
- ✓ The else block is not executed.

▤ num1 <= num2

- ✓ num2 is returned.
- ✓ num2 is the maximum number.
- ✓ If both numbers are equal, num2 is returned.

✦✦ Summary

- ✓ Reference parameters allow a function to modify variables from the calling function.
- ✓ An if-else condition can be used to find the maximum of two values.
- ✓ A function can return the result of a comparison.
- ✓ Functions can be combined by passing the return value of one function to another.

Thank you 😊